



New way to remove CO₂

MIT engineers have come up with a new way to remove carbon dioxide directly from the air. <https://dgit.in/nov19-73>



Disney's late night show

Disney's streaming service, Disney Plus, is getting a muppet late-night talk show host. <https://dgit.in/nov19-74>

ASUS ROG Zephyrus G GA502

A cheaper alternative to the Zephyrus M GU502

The new entrant, ASUS ROG Zephyrus G GA502, appears to be a watered-down version of the original Zephyrus M GU502. Let's see how it holds up.

Build and design

The Zephyrus G GA502 is a well-built and well-designed product. It sports a metallic top cover with a classy brushed finish. For a 15.6-inch gaming laptop, it's moderately slim (0.8 inches) and surprisingly lightweight (2.1 kg).

Opening or closing the display reveals almost no sign of flex in the lid, which is impressive, given the slimness of the panel. The 15.6-inch anti-glare display is surrounded by thin bezels on three sides. The tall bottom bezel houses just the dual-array microphones. On the whole, the Zephyrus G GA502 looks subtle without looking uninteresting.

Display, audio and IO

The 15.6-inch display is a 120Hz IPS LCD unit that's bright but not as colourful as the Zephyrus M GU502. Primary colours look mostly neutral for browsing, and playing games, but they tend to appear a bit washed out on occasion.

Sound from this gaming laptop is surprisingly loud and clear. There's ample stereo separation as well, however, you might want to get yourself a proper pair of headphones for gaming.

The Zephyrus G GA502 is not shy to offer ports for connectivity. On the left side, it sports a round-pin power port, a LAN port, a full-size HDMI 2.0b port, a USB-A 3.2 port, a USB-C 3.2 (Gen 2) port with DisplayPort 1.4, and a 3.5mm audio jack. On the right, it features two USB-A 3.2 ports and nothing else.

Keyboard and touchpad

The keyboard on the Zephyrus G GA502 is adequately comfortable for



Price
₹99,990



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typing long emails and documents.

The same goes for gaming. The keys have good resistance but could use a bit more travel. A dedicated row of keys lets you control volume, mute the microphone, and bring up Asus' proprietary Armoury Crate app. The one thing the keyboard misses is RGB lighting.

The touchpad on the Zephyrus G GA502 is a sufficiently large precision unit. You can perform multi-finger taps and swipes without needing a third-party driver or utility. The two click keys beneath it are fairly easy to press.

Performance

The ROG Zephyrus G GA502 features an AMD Ryzen 7 3750H combined with 16GB of RAM, upgradeable to 32GB. Complementing it is an Nvidia GeForce 1660 Ti GPU. There's a single Intel 512GB PCIe NVMe solid-state drive attached to the laptop along with space for one more PCIe NVMe card.

Everyday performance of the review unit was expectedly good. We were able to run multiple instances of numerous applications across more than two virtual desktops without any noticeable signs of lag or stuttering.

The review unit fared rather well on CPU and GPU benchmark tests. On PC-

Mark 8's Conventional Creative test, the Zephyrus G GA502 scored 3783 points. On 3DMark's Fire Strike and Time Spy, the Zephyrus G GA502 bagged 9820 and 4242 points respectively. The performance on the Zephyrus G GA502 is not better than the Zephyrus M GU502, but it comes quite close.

Coming to gaming, Doom ran at an average frame rate of 73 frames per second on High but jumped to 88 when switched to Medium. Playing video games was mostly pleasurable. However, the device was noticeably loud while playing games.

The Zephyrus G GA502 sports a four-cell 76Wh lithium-ion internal battery, which, on our tests, fared decently. On our standard battery benchmark test, the review unit lasted 3 hours, 1 minute. Charging the laptop back up from near-zero to full takes a little over one and a half hours.

Bottomline

If you choose to get this watered-down version of the Zephyrus M GU502, you won't be losing out on too much in comparison to how much you're saving. You'll end up with a (slightly) less powerful CPU and GPU but it won't make a vast difference to your gaming experience, especially if you're a casual gamer.

However, if getting some extra frames per second is important to you (along with a more colourfully lit keyboard), choose the Zephyrus M GU502 over this. If not, get the Asus ROG Zephyrus G GA502 without any hesitation.

—Vignesh Girdharan

SPECIFICATIONS

PROCESSOR: AMD Ryzen 7 3750H | GRAPHICS: NVIDIA GeForce GTX 1660 Ti | MEMORY: DDR4 2400MHz SDRAM Up to 32GB | SSD: M.2 NVMe PCIe 3.0 x 2 256 / 512GB | DISPLAY: 15.6-inch Full HD, VIPS-level panel, 60 / 120Hz | KEYBOARD: Backlit chiclet keyboard | AUDIO: 2 x 1W speakers with Smart AMP technology | WEIGHT: 2.1kg

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